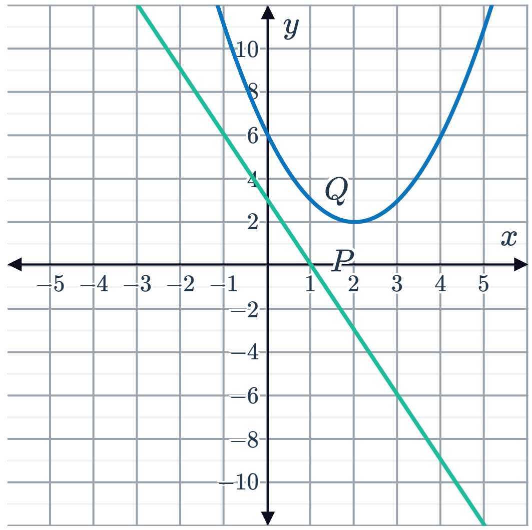
Compare function types

1

a Function P is a line and function Q is a parabola.

b

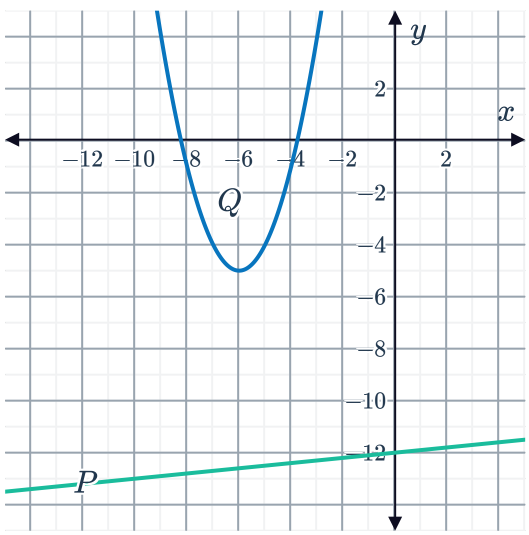


c Function Q

d Functions P and Q are decreasing.

2

a



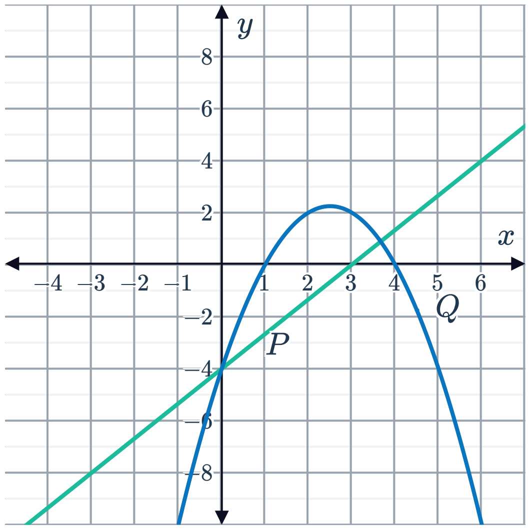
b Zero times

c Function Q

d Function Q

3

a

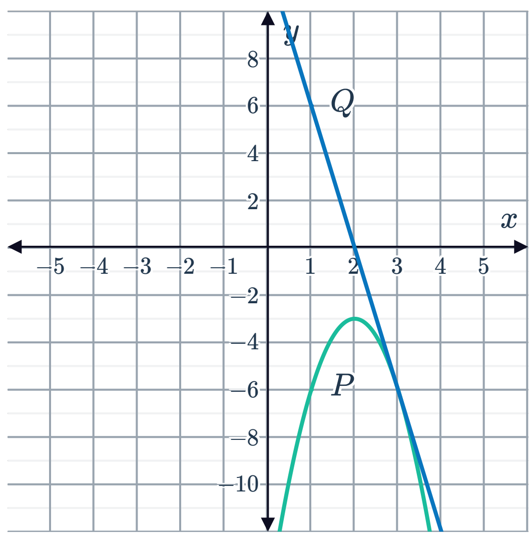


b Twice

c They are equal at $x = 0$.

4

a



b Once

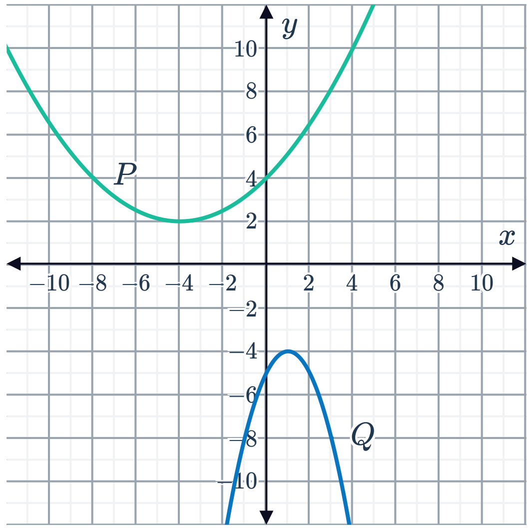
c Function Q

d Function P is increasing and function Q is decreasing.

5 Function P

6

a



b Function P

c Function P

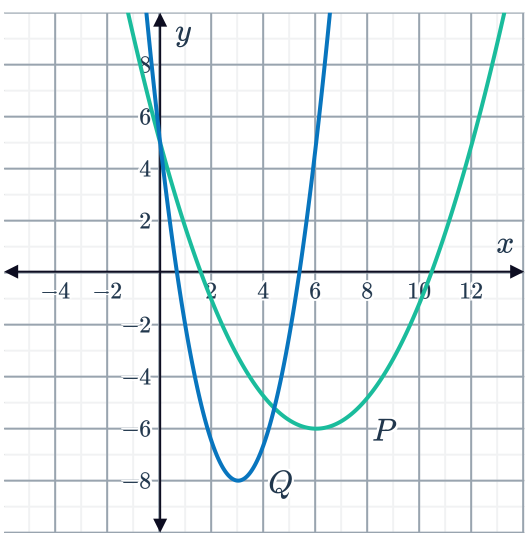
d Parabola P and parabola Q are increasing.

7

a Function A **b** Function B **c** 2 **d** 0

8

a



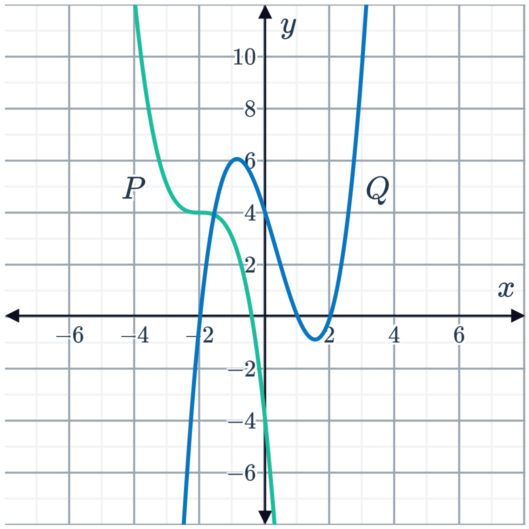
b Parabola Q

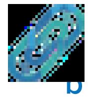
c Twice

d Parabola Q

9

a





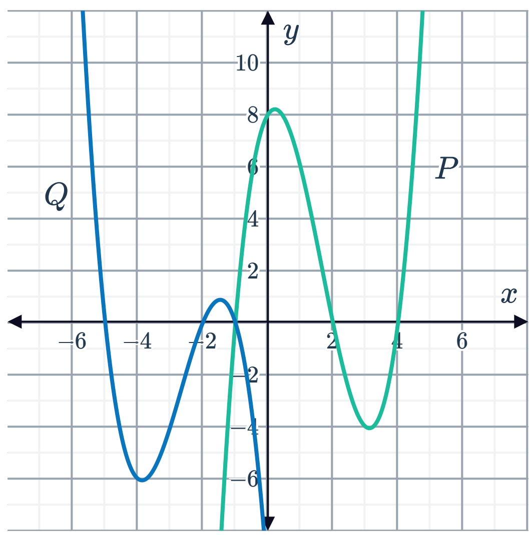
b Cubic function Q

c 1

d 3

10

a

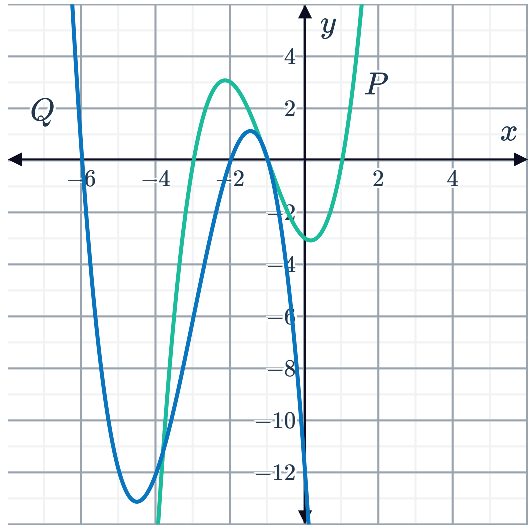


b Function P is decreasing and function Q is decreasing.

c As x approaches ∞, function P approaches ∞ and function Q approaches $-\infty$.

11

a



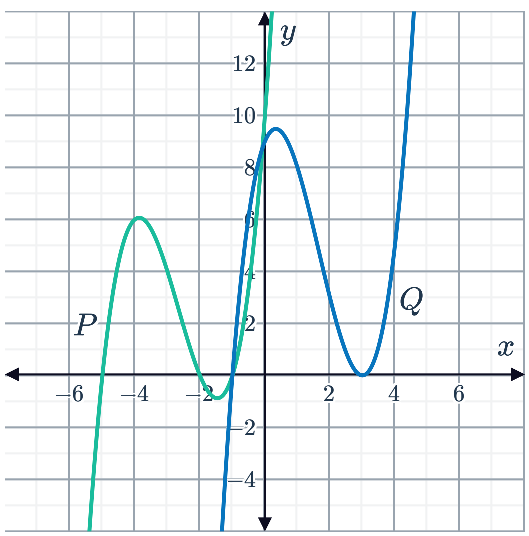
b Function P

c Function P is increasing and function Q is increasing.

d As x approaches ∞, function Q approaches $-\infty$ and function P approaches ∞.

12

a



b Cubic function P

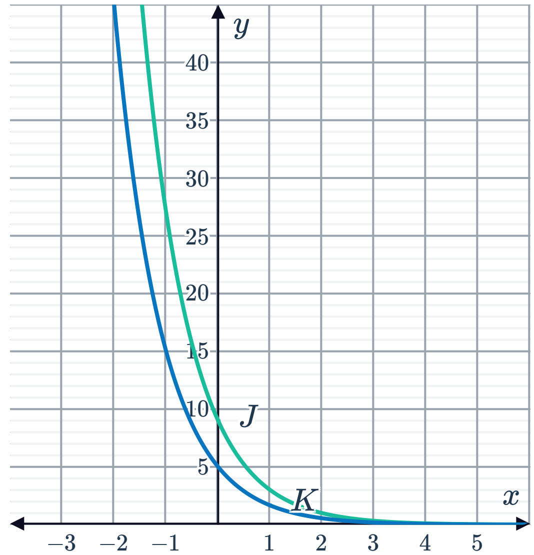
c 3

d 2

13

a Decreasing

b

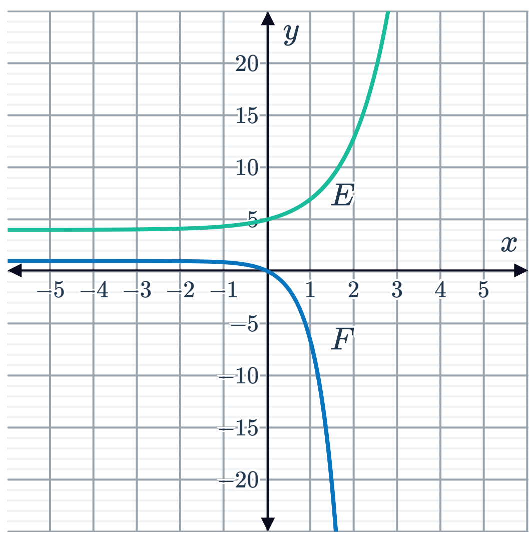


c As $x \to \infty$, functions $J,\ K \to 0$.

d Dilate $y = 3^{-x}$ vertically by a factor of 5.

14

a



b Zero times

c As $x \to \infty$, the function $E \to \infty$ and function $F \to -\infty$.

15

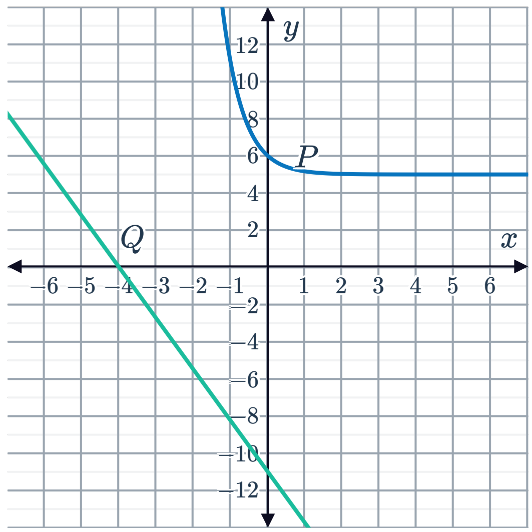
a Once

b Function P

16

a P is an exponential function and Q is a linear function.

b

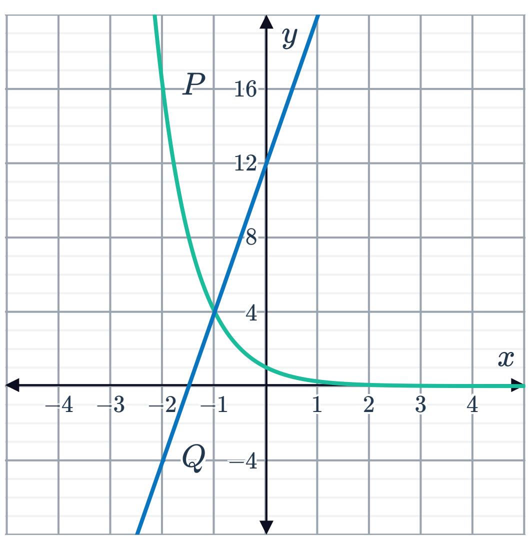


c Function P

d Zero times

17

a



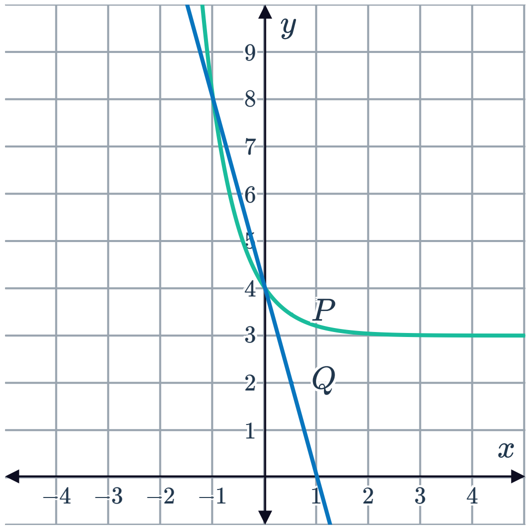
b Once

c Function Q

d Function Q

18

a



b Twice

c Function P

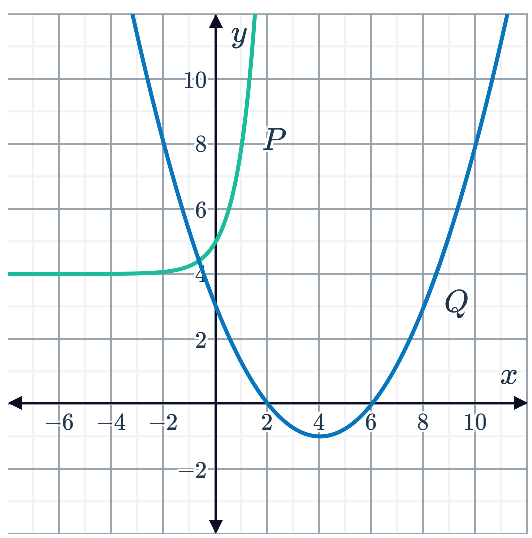
19

a Function A

b Function A is decreasing and function B is increasing.

20

a



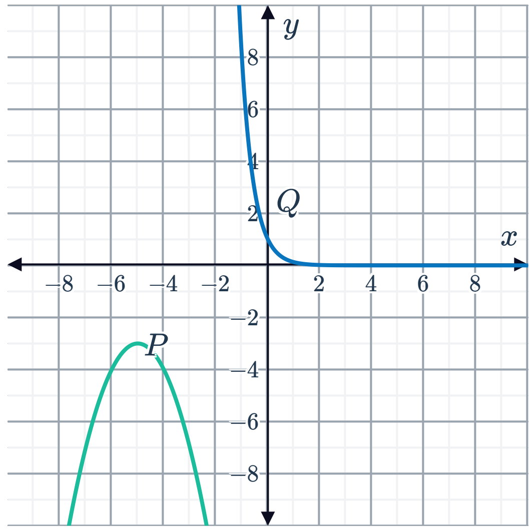
b Function P

c Function P is increasing and function Q is decreasing.

21

a Function P is quadratic (parabola) and function Q is exponential.

b



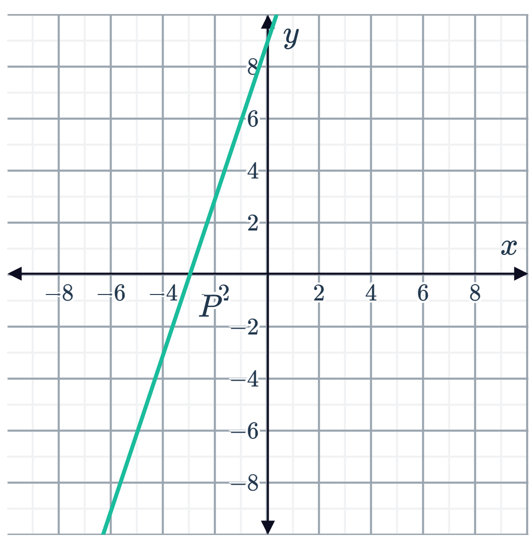
c Zero times

Applications

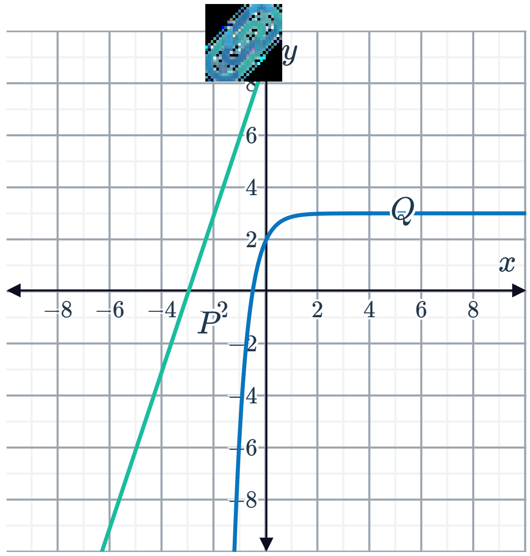
22

a Function P is linear and function Q is non-linear.

b



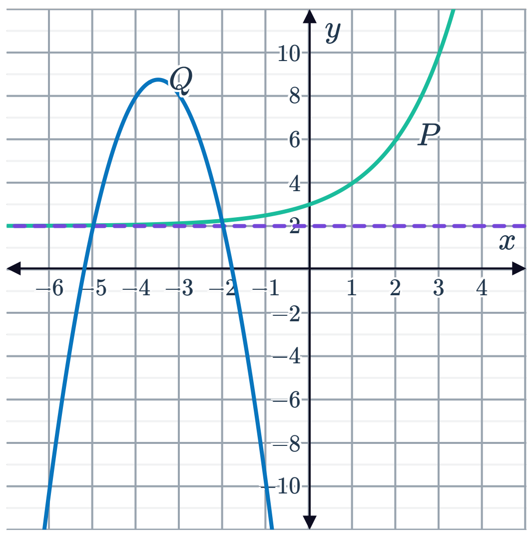
c



d 0

23

a



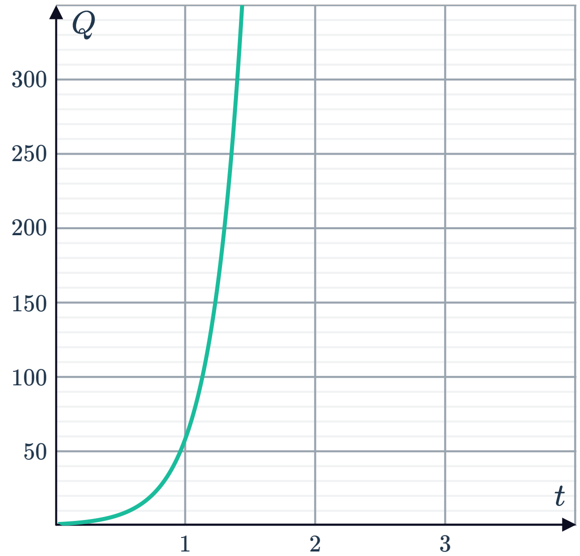
b As x approaches ∞, the function Q approaches $-\infty$ while function P approaches ∞.

c 2

24

a The population of bacteria J increases faster than the population of K.

b



c